ActInf Textbook Group ~ Cohort 3 ~ Meeting 8 (Chapter 4, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 8 (Chapter 4, part 1) | 55:07

Hello, it is March 15th, 2023. We're in our first discussion of Chapter 4 in Cohort 3,
and there's a lot of great questions. First, is there any general comments
or reflections that anyone wants to share about Chapter 4?
Yeah, just previously people were reflecting on this kind of accuracy, complexity, trade-off,
and how much and what maths to show. Ali?
Yeah, as Jonathan and Francois also mentioned, this chapter is pretty dense,
probably one of the most difficult chapters in the whole book.
But in my own experience, the step-by-step tutorial paper is extremely helpful when reading this chapter
because some of the missing gaps in the explanations and some of the, in my opinion, necessary details
that are somehow, for some reason, have been omitted from this chapter
can be found in quite helpful detail on that paper.
So, yeah, I definitely highly recommend reading that one.
Oh, that's great. I will do so.
The thing is, I know that the step-by-step paper is only for discrete time,
but this Chapter 4 also goes to continuous time.
Is there a similar resource that you could recommend for...
Raphael Bogach's paper on tutorial on free energy principle
somehow does the same kind of, not simplification,
but kind of pedagogical walkthrough through the continuous time active inference.
So, both of those papers can be seen as complementary to each other,
one for discrete and the other one for continuous time situations.
Thank you. Added the link here.
Any other general comments?
We had Chapter 1, Introduction.
Then, we walked on the low road and the high road to active inference.
And now in Chapter 4, we're at the generative models of active inference.
So, if it's an active inference generative model, it's active inference.
If it's something else, it's something else.
A focus in this whole textbook and field is about how time is represented.
And those who have done any kind of dynamical systems modeling
will be familiar with a lot of the fundamental differences and challenges.
For example, continuous and discrete representations of time,
and how to discretize time, and so on.
So, Section 4.2.
From Bayesian Inference to Free Energy.

The preceding two chapters with the high and the low road
were outlining important connections with active inference and other paradigms.
And especially the low road helped us look at the Bayesian brain.
Starting from the bottom, starting from the how.
With respect to the high road, this is where we bring in notions of cognitive entities
and the kinds of bounded inference or bounded rationality
that support self-organization and persistence and adaptive decision-making.
This chapter recaps Bayesian inference
and specifically the variational approach of Bayesian inference.
That is what is going to connect generative models,
which are how active inference models are specified,
to the concept of free energy,
which is going to be shown to be kind of like a criteria or an imperative
that's used to fit or describe different generative models in the data.
This section is more technical than the previous chapters,
appealing to a little linear algebra,
differentiation in calculus,
and the Taylor series expansion.
Some of those topics are in the appendices,
and even the points that people like on the structure
and what would have been helpful to know when.
Those are all helpful comments to write or to surface.
There's no one single linear layout that would have been perfect for every single case,
but even just like sharing what would have made it helpful for any given person
or what detours or cul-de-sacs they went on,
that's all really useful.
Those who do not want to delve into theoretical underpinnings may skip this chapter.
So soon.
Bayes equation in 4.1.
Does anyone want to just bring up anything or describe it or call it like they see it?
Let's see if we have the natural language description.
Let's see if we have the natural language description.

Let's see if we have the natural language description.

Let's see if we have the natural language description.

see like how it can be read out. But a key issue is that whether you're summing over discrete opportunities or integrating over continuous ranges, these sums or integrals are very computationally challenging or even intractable. The approach of variational inference is to convert this potentially difficult summation or integration problem into an optimization problem. So instead of just like one mega plug and chug, crunch the numbers and get one outcome at the end, there's going to be a incrementally optimizable approach that we can iterate on with known computational complexity that under some or potentially a broad range of settings will bring us closer to the best reasonable compromise answer. And so we've exchanged this vast one-shot calculation into something that can be incrementally through time on a real computer optimized.

To understand how this works, we need to appeal to Jensen's inequality, which says that the log, note that's not the log squared, that's a footnote subscript, the log of an average is always greater than or equal to the average of a log. And figure 4.1 shows that. So does anyone, math professor or otherwise, want to give like a thought on this figure 4.1, what is happening and why is this relevant for how we're going to be continuing on with Bayesian equations?

Maybe not a particularly deep thought, but so this is because of the second derivative, the fact that this is kind of curving down, the steepness is getting less that this holds true. I created a little animation that I can link to looking at sampling along x and then comparing the logarithm of the expected value with the expected value of the logarithm, just to get a bit of an intuition as to why this holds true. But I think that this image is pretty nice. This sort of shows why it is. You can imagine that any function which is curving down like this is going to have this property and any function which curves up is going to have the opposite property. I can paste a link into the comments of the little animation that I created there.

Awesome. And can you maybe add one piece? How is this connected to Surprise? Or how does anybody see Jensen's inequality maybe as leading to the way that we're going to talk about Surprise?

So only that this is going to give us a bound on Surprise because Surprise is the average of the, of x log x or p of x log p of x essentially. That we're averaging the amount of information that we will gain on making any particular measurement. It's the average information that you'll gain. And so that's already an expectation and it allows us to do an expectation of x and take the logarithm of that, which gives us a bound on the, on the Surprise.

Thank you. It's definitely like some words that fly by and to hear it many times is good.

It does turn out that this property of logs, and there's going to be multiple properties of logs that
come in to help us in different ways, like the ability to take two numbers that are multiplied
inside of a log and make it into log of a plus log of b and so on. But this curvature property
is going to help us talk about expected Surprise. Okay, I'm going to bring up the...
Awesome.
Could you connect that to any sampling or, or like what are we being surprised about? Something on the
y-axis, something on the x-axis, where are the data and so on?
And so, so this is just a normally distributed x here. So there's some probability over x. And then the,
what you're looking at, I guess in the, in the, I've got to think about this now.
No, I don't want to say, say without thinking about it. But essentially here, this is just taking some
expectation over x and calculating the logarithm of that expectation and the expectation of the
logarithm, just sampling. So starting on the left hand side with, with, with just one sample,
and then increasing to 100 samples and just seeing how the relationship between the two,
those two points is. Cool. That's, that's all it's doing. Cool.
Now we're going to take advantage of that property. So we return to equation 4.1. So this was the one
with the intractable sum or integration. We're going to return to 4.1 and multiply by an arbitrary function
q. So we took this line with big challenge and we said there's some arbitrary function q,
and anything divided by itself is one. So this is equivalent to multiplying by one on this left side.
And it turns out that q is going to be like this variational distribution from a known family
with easily optimizable characteristics that we know we can control.
Now, there's going to be a log taken on both sides.
So q of x over q of x, which is equivalent to just one. And then there's going to be
the log taken on both sides.
We can now interpret this expression as an expectation, fancy E, not E cubed, that's just a footnote again,
E of a ratio between two probabilities.
So here's the p of y of, this is the joint probability distribution of hidden states and data.
And now this allows us to construct joint distribution over the one we control.
Natural log of that, natural log on the outside, is always greater than the expectation of the log.
Log of an average is always greater than or equal to the average of a log.
So this value, this is again, this is the part that we want to solve.
And this is another way to rewrite it.
That is strictly bounded via Jensen's inequality to this value,
which is defined by triangle, the negative variational free energy, as a function of
q, the distribution we control, and y, the data.
The smaller the free energy, the closer it is to the negative log model evidence.
Especially because the log has these like somewhat strange seeming properties, like it's
shoots off down to negative infinity over here and so on, like, and there's negatives and positives.
So, you know, take it slow to see what each line is actually telling us.

Um, but this is the, um, standard from statistical physics, what is called the negative variational free energy.

When free energy is small, it's close to the negative log model evidence.

So, if the log model evidence were really high, like, then the log, because of it's always going up and to the right,

but slowing down in how it does that, it means you're also monotonically getting higher model evidence.

And also, the natural log of p of y is, um, how that's shown.

So, let's go ahead and see what we're going to do in the next episode.

Any thoughts on equation 4.3?

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4.3 provides some intuition for the roles of the free energy and the q distribution.

Two quantities that were difficult to compute without the variational approximation.

So, the expectation about the distribution that we're getting to control and parameterize,

the expectation of the likelihood, which is natural log of p,

of the joint distribution between the hidden states and the data,

is the negative variational free energy as a functional of q and data,

plus an expectation of the model likelihood of q.

And then, there's some rearranging, bringing the f onto this side,

and then, using the, um, properties of differences of logs,

the value of the matrix to get us to this representation.

Variational free energy with two components.

The raw model evidence of the data,

the distribution of the data,

and a KL divergence between the distribution we control about x, hidden states,

and this

p distribution of x given the data.

Let me just add in your comment there.

So, Jonathan wrote,

um, the entropy is the average surprise.

Surprise is not a single measurement, not itself an average.

It is simply the probability of negative natural log of p of x.

Thank you.

The KL divergence is defined in the second line as the expected difference between two different log probabilities.

This is a measure, it's not a proper measure,

of how different two probability distributions are from one another.

So, when there's no divergence between the distribution that is under control,

and the sort of true distribution that we'd want to estimate,
then the relative variational free energy has been minimized.
Because the log model evidence is just,
like, whatever it is, it is,
the value of the energy that is with respect to being able to change q.
Which is what makes the free energy an optimizable quantity.
Because for a q where you know which knobs can be tuned,
and there are methods for the incremental optimization,
like on smooth optimization landscapes and so on,
then this divergence can be very directly tackled.
And this is just like, can be kind of just, um,
left unconsidered, because it's a constant with respect to the same data.
Assumed.
What is?
It can be assumed.
Yeah, assume that whatever it is,
with respect to changing the knobs on q, it doesn't matter.
Because q is not included in this term.
q only shows up here.
So, yeah.
Okay.
Yes, definitely a few, um, tricky,
simple seeming, but very ineffable transformations with logs and the Bayes equation
that get us to a form for variational free energy,
which if we remember back so long ago,
is going to help us bound surprise.
And there's going to be like a special case that doesn't necessarily ever get realized where
the free energy, like the bound divergence goes to zero.
But in any situation where the distribution p is maybe even of a different structure than q,
there's always going to be some divergence.
However, by reducing that divergence, we will be doing the best optimization we can.
Michael or anyone?
Okay.
Okay.
Yeah.
These are some, some of the most pure mathy pages in the textbook.
Sorry, my mic's on.
It's all good.

No, it's, it's true.
It's like, where did active inference go?
But this is totally in it.
But these several pages, which may be in a, um, undergraduate statistics course could have been unpacked into several lectures and homeworks are, are very fundamental, very much brought in quickly here to get us to this variational free energy.
As someone who is not the mathematician, but, um, uh, spoke it at one point and, um, is grappling with this question of how do I think about these objects in order to speak to others about it, uh, earlier in, at the beginning, there was this, um, uh, I was thinking about the original, uh, uh, formula as inputs to a process that then, uh, there was a calculation being done.
And that, that input was one of the false characterizations that, um, that it's an inflow.
So it's not a single, but it's a, it's this idea of there's a continuous.
And to think of surprise as also as a, as a living, flowing thing is also one of those kind of hard to grapple with that we're not talking inputs to, to, you know, but rather flow to flow to, and that there's the, um, the hearing the, the, the variability, right?
What's not explained, um, is, is the, is, is, is the intuition that's hard to cultivate for those of us who these, these, these formulae just kind of paralyzes and, and it's, and trying to translate what are we learning here, um, is, uh, that's where I'm, I'm realizing my existing language doesn't capture what the mathematics is describing. So I'm contributing my, my confusion. I hope articulately.
Thank you. Um, just to Jonathan's comment and then back to that. So, um, Jonathan, yes. However, you, whatever format you're taking your notes in there's, there's always going to be a way, um, in the equations themselves, you could always tag anything.
Like you could add descriptions or just notes or other unpackings or other links in the equation section, or we can go and modify or revive the math learning group with any different notes that people have on, um, notation on different resources to look at or like kind of chapter wise descriptions of math.
Um, hopefully not to confuse or, or devalue this too much, but I hope to bring even, um, more clarity. Ultimately, if you check out live stream number 52, especially the final hour of 52.2, you'll hear about Lance DaCosta sharing about the free energy divergence is just one discrepancy measure that can be utilized in pursuit of a unified imperative for on the inbound signal processing and on the outbound control theory. So signal processing alone is just passive inference.
Control theory alone essentially has to presuppose knowledge of the world, or at least it's less considered with it. Active inference is considering the inbound and the outbound and free energy energy. It turns out is an extremely convenient discrepancy measure, which is why even before all of the two thousands and all of that free energy and variational auto encoders, Bayesian statistics were already in heavy use because free energy as a bound on surprise in Bayesian statistics is just useful.
But it's not the only discrepancy measure, but in the variational setting where we're learning it here, it is the divergence or the discrepancy measure. But thinking about this less as like we're getting

to active inference from free energy, what is free energy? Put the horse before the cart, the cybernetic loop, and the signal processing inbound, control theory outbound, unified imperative first, and then these are going to be different useful approximate Bayesian calculations, but not the only calculations that can be used for modeling in that cybernetic setting.

Like just one note, then Ali, the KL divergence describes the informational distance, not exactly distance, but basically between two distributions. But he described another distance measure, the earth movers distance, how much earth you need to move from one distribution to make it resemble the other distribution, since they both have an

area under the curve, let's just say of one. And those are different measures. And they have different properties. There are situations where one might be appropriate or not. So one could actually do active inference without free energy calculations, there could be a different discrepancy calculation. It's just that this is a go-to choice for the same reason why free energy calculations are not arcane, they're workhorses in variational autoencoders, and so on. Ali?

It's just related to what you mentioned. Two related pieces of literature just came to mind. One is this well-known paper, A Tale of Two Densities. Active inference is inactive inference, which contrasts two interpretations of active inference, namely structural representation and inactive inference representation, which they basically somehow lean towards the second type of interpretation. And the other one,

which is not so well known, is a chapter from this book called Measuring and Modeling Persons and Situations by

Chapter 18 from this book by Adam Saffron and Colin D. Young, namely Integrating Cybernetic Big Five Theory with the Free Energy Principle.

So this is also a really helpful synoptic comparative analysis between different cybernetic and control theories and active inference. So yeah, I just wanted to mention those two papers.

Thank you. So again, not to devalue this whole variational sojourn, but this is the variational approach. That's why we went in with a Q.

Another approach could have been, you know, this is a really intractable sum. We're going to sample from it with Monte Carlo simulations.

And we're not going to do variational distribution at all. So we can still talk about active inference even without this variational technique. So it's like, this is not, this is not the essence of active.

However, variational Bayesian methods are incredibly powerful and scalable on modern computers, including in some places where sampling is like going to be costly or impossible.

But it's really interesting to think about everything technical is a modeler's degree of freedom.

Once one has made the decision to have a unified perspective on perception and action, that is the active inference turn.

And now these are modeler's possibilities, sampling variational Bayesian methods and other approximate Bayesian computational methods.

And those paths are not always obvious because it's like the variational free energy. We're using variational methods, but it's not the only method that could have been used.

But then it's like, what is most commonly shown.

Um, so long, like kind of path to return to, we got to variational free energy, as has been reached by other Bayesian statisticians.

And from that, from that from that, from that from

distribution that we control and the one that we would want to have no divergence with

is literally, that is the KL divergence, is a difference of these two logs, which is also

why some of those more fundamental properties and logs were being brought up.

Okay, 4.3. To calculate free energy, we need three things. Data, a family of variational

distributions, and a generative model. So do organisms minimize free energy, calculate free

energy? No. Data, a family of distributions, and a generative model do. Again, hard to sometimes

see the rippling consequences, but this is a pure instrumentalist expression. So cells do not

minimize free energy. They don't calculate free energy. The models do.

There are going to be two families of generative models, and the exact form of free energy is

going to be slightly different for these two. It's kind of like vertebrates and invertebrates.

It's like there's a lot of decisions down the line, but this is like some of the broad taxonomic

divisions in the generative models. The first is going to be dealing with discrete state spaces

and discrete representations of time. And the second is going to be dealing with continuous variables

and continuous treatment of time.

Before showing what those generative models look like, figure 4.2 gives some fun intuition

for this Bayes' graph diagram format. So circles represent random variables.

So those are things about which we hold beliefs. It could be a fixed or a learnable parameter or

stochastic variable. And squares are probability distributions that are like connector pieces

because they describe relationships. And so these are laying out situations where it's like,

here x is upstream of y. Here x is upstream causally of y and w. Here there are two causes upstream of y.

And here is a chain of causation. And so in this directed Bayesian graph setting,

these are sometimes called causal graphs because they summarize the causal influence of one variable on

another. So correlation is not causation. And then it gets a little bit mingled because these are

called causative graphs, but this could be living in this region and this could be having this disease.

But this is just a model. So that doesn't mean living in the region is the mechanistic cause of having

the disease. And that's why we're always talking about evaluation of different possibilities.

Michael, how completely do these four representations of causality cover the known forms of influence?

Someone can give a thought. My first answer would be, these are like, this is like subject verbed,

subject verb object, you know, subject, you know, these are like the clauses that influence networks

are constructed from. So in the ultimate, in the ultimate modular sense, like this captures it all.
A influences B. But then in the real setting with like interacting and emergent causation,
these get stitched together in complex ways. And in fact, the generative models that we're going to
look at are going to be using this kind of a visual grammar to describe how different variables
influence each other. But it's an interesting question about what forms of appreciation
influence and control exist and how those are represented with Bayes' graphs.
Okay, a little bit of unpacking on the, what they are talking about with the disease and the diagnosis
and all of that. We get to figure 4-3. These are going to be some core visualizations.
The top part of the graph is going to show a partially observable Markov decision process or a POMDP.
And that's going to be in discrete time. And the bottom part of the graph is going to be displayed
and juxtaposed to highlight the structural similarities. But then there's going to be some very,
very important differences as well. And that is also going to be unpacked in chapters 7 and 8,
which are respectively about the discrete time and the continuous time settings.
So these figures you will see in many active inference papers. Thomas Parr in the recent book
Stream gave a little bit of a historical account. He emphasized that earlier on, like pre-2012,
there was a lot of continuous time modeling, like with eye movement and the reflexes and motors.
People then developed the discrete time, discrete active inference, because it is classically used to
model decision-making, categorical and discrete decision-making. But these two kinds of models
can also be hybridized, and that's called the hybrid model. So here, again, just to see it one time,
but we'll see it again and again. It's like a rake on the bottom, and then there's an action selection
part on top. Discrete time, continuous time. And then even the hybrid model, it's like here's the big rake,
top-level model. And then where there's just a little leaf here, that has a whole continuous time
generative model grafted on. So again, like no GM, no generative model is like the act-inf generative
model. It's like a grammar or a toolkit for comparing and evaluating portfolios of structurally
different or parametrically different models. So it's not like there's one way to look at attention,
there's not one way to look at memory, there's not one map for a territory, and so on. But there's going
to be just like many kinds of intuitive graphical operations that this format is going to help us talk
about. Like here is the past, present, and the future. And you say, well, could we have like three
more? Could we go four time steps into the future? You could. Or could the observations be smell and taste?
Yes, they could. Or could there be 10 policy options? Yes, there could be 10 policy options.
So it's like just a skeleton that is going to be used and modified and kind of remixed in many,
many ways. Does anyone have any like thoughts or descriptions on figure 4.3? What's something,
whether they've read on past this section or not, like what have they found important in 4.3?
And then let us like dwell on 4.3 a little bit, look to the questions, and then next week in our
second discussion we'll continue on from here and look at more questions.
So earlier the step-by-step was brought up.
And this figure, which I'm sure we have pasted many times into our textbook group coda,
is working with that same graphical grammar of the Bayes graph. On the left, we have a prior

that is seeding a hidden state S, and then the hidden state S is emitting an observation O.
You can set S and generate O, that's generative modeling, generative AI. You can also have O and update S, that's the recognition density. So A, which is called the emission matrix, is the tail of two densities. Because you can have a hidden state and generate synthetic data from it, that's generative. Or you can have an observation and use it to update your hidden state, and that's recognition. So this is the tail of two densities in the static setting. So this is like a single image and the image classifier label. And so generative AI is going from, draw me a cat, to generate the pixels. Whereas previously, there was only really recognition AI from the pixels to, oh, that's a cat. But it's static in the top left. The bottom left is going to be that same motif, tail of two densities. But now, S has a subscript for 1, 2, and 3, which are three time steps.
And so S is changing through time. The way S changes is through B, which is called a transition matrix. So if there's like two states in S, you know, is it could be on the top or the bottom.
There's going to be a two by two transition matrix, where the diagonals are like staying where you are, and the off diagonals are moving. And that is going to be the describing the time evolution of the hidden state. And then at each time, there's the tail of two densities and the emission of an observable.
And the A matrix can be sharp, which is like having kind of ones across the diagonal.
So you can like see the hidden state without error. Or in the other extreme, A could just be like, kind of like a wash, at which point the observation wouldn't tell you anything about the hidden state.
Or in the middle is kind of like a semi-blurry A matrix, where there's information on the hidden state from the observation, but it's not unique. For example, a positive test result doesn't mean you have the disease. And that's a classic like Bayesian statistic. Oh yes, thank you. A does not have to be square matrix. I'm using square examples. But yes, the dimensionality is like a huge thing to make sure that especially in complex models that the dimensionality like works out.
So this is passive dynamical perception. We're getting temperature readings at time one, two, and three. And whether we have a perfect thermometer or whether we have like a kind of noisy thermometer, we're going to be making our inference about the hidden state, the true temperature in the room through time. And then B describes how the temperature in the room changes through time.
Whereas A is time independent, and it describes the mapping between thermometer readings and temperature.
This is passive inference on the left. Michael?
Again, going to a different language, but in management, traditionally, management has been unidimensional. So for example, you might say we need to manage the dollars being produced in this process or the fishery, the single species in this fishery.
And that one of the challenges has been to look ecosystemically, instead of through a unidimensional variable that is optimized, at the interplay between all the variables. And that's what I'm witnessing here in a different space and sense, this kind of multivariate management of the interplay between all the factors. That's my own shorthand for what I'm thinking, I'm sensing.
Awesome. Yeah. The observations could be our triple bottom line, and then the hidden state could be the

unobservable health of our organization. And so we're not just doing this like naked optimization on an observable, but we're kind of inferring from the observables into an unobserved space. And then that's the space that we're describing the dynamics on.

Now, all of this has been passive, but this paper goes step by step. It introduces static perception and dynamic perception. Here's where action comes into play in active inference. Pi is policy. So there's a slight difference between affordances, which are the actions that can be taken in a moment, like up, down, left, right on your joystick, and policies, which are sequences of action over a given time horizon. So with a time horizon of one, your affordances are your policy repertoire. But for a time horizon of two is up, up, up, down, up, right, you know, and so on. So pi describes policies. So that's sequences of actions that can be taken for the length of the time horizon, which is a modeler's choice. So pi describes which policies can be taken. How are policies evaluated? Well, in reinforcement or reward learning, you would have utility function that ascribes a utility value to different policies. In active inference, we have expected free energy as an imperative that guides policy selection. Here, free energy G gets input from C, which is our preferences or our expectations. And so this is that fun twist with what we expect slash prefer is like in reward learning, we would say the body's rewarded by having a homeostatic temperature, survivable temperature bound. In this kind of preference based active inference, you say the body expects to be in homeostasis. It is surprised if it is not. And it takes actions to reduce the divergence to minimize and bound its surprisal. And so action policies are selected, not because they have a high estimated utility, though that can be the case, but rather because the policy reduces expected free energy by aligning observations with preferences. It's also important to note that policy and action selection intervenes in the B. So this is changing how hidden states change through time.

And then this bottom version, which is not really even gone into much in the textbook, they're adding another variable that describes like a temperature, a precision on policy, where in the absolute zero temperature, high precision setting, if there's two policies and one is a little bit better, you'd want the one that's slightly better every time. In the neutral temperature, if it was 51, 49, you'd want to select that one 51% of the time. In the high temperature, you would erase the differences between policy choices, and you would be selecting equally without respect to the mentioned criterion. But it's a little bit of like a modification. But also, it sets us up for some of the incredible elaborations that we're going to see within this visual grammar. So we're going to see hyperpriors on different variables and learning rates. And it's like, this is like a circuit board. There's no single circuit board. Because you can wire things up differently, you can connect and say, well, but what if there was, you know, L that influenced C? It's like, you can make that generative model. And with Bayesian statistics, you can evaluate and juxtapose models that have radically different anatomies. So it's like, these are circuit board designs, or like Lego architectures. So there's not going to be one GM, even for a very narrow system of interest, any more than there would be just a single circuit board.

All of that, and all of step by step, is in the discrete time setting. So past, present, and future,

how things change through time, how hidden states emit observations, how the initial hidden state is set with a prior, and how policy selection intervenes in hidden state through change, change through time, and how preferences shape through expected free energy minimization policy selection.
In the continuous time setting, it's a little bit different. We're going to come to it.
It turns out that there isn't explicit prediction of past, present, and future time steps, but rather, generalized coordinates are used, which is to say like position, velocity, acceleration, and so on.
Generalized coordinates are used akin to a Taylor series expansion, such that a smooth and differentiable function is proposed that makes continuously interpolated predictions without explicitly mentioning discrete moments and giving explicit predictions for them.
So there's some pros and cons in different settings and so on, but we return to this discrete, continuous time distinction repeatedly. Fundamentally, it's a modeler's choice,
often guided or constrained by the kind of data that one has. And that's going to come up again and again. They're laid out here in figure 4.3 to illustrate their structural similarities.
In both cases, we have the kind of rake or this like E facing down with the hidden states changing through time and emitting observations. Caveat for, you know, the way that it's a little bit different, but basically that's what's happening. And then the upstairs is the policy selection.
And that's what we see in step-by-step. The rake is just static or dynamic inference.
That's signal processing. That's the Kalman filter.
And then the upstairs is policy selection. That's control theory.
Control theory upstairs, signal processing downstairs.
All of it is being included in the same generative model
so that we can have summary statistics that are optimizable for our generative model,
like free energy, so that we can have a unified imperative for perception and action.
Let's look to what questions we'll come to next week.
All right. Equation 4.10. So equation that we haven't gotten to yet, but yes, there are more equations, including some subtleties with subscripts in bold. And so these are definitely ones to have good readings of, like, a lot of different questions and like, why is that line doing that?
All of this. We'll come back to it.
The expected free energy is minimized by selecting those observations that cause a large change in beliefs.
We'll come back to it.

Okay. A question of variational inference.
And also we can use the Coda LaTeX pack to make it render nicely.
Equation 4.10.
Same here.
And some more questions on equation notation.
So thanks for these very careful comments on the notation there.
Awesome.

So yeah, a lot on the equations.

And it's like we can always return to the math learning group.

We can set another time.

Because there's math learning questions

questions to help increase the approachability of this entire corpus.

And then there's also some really specific technical questions where it would be awesome to just have, like, here's what the squiggle means.

And so those are like two different regimes of attention within the math learning mode.

And the math learning mode is just one mode that we're in overall.

But both those accessibility and rigor components of math, hopefully, like, if people are motivated to somehow work on that, there's like a lot of opportunity there.

Any final thoughts?

Or we will return next week in our second discussion on 4.

All right.

Great.

Thank you all.

Farewell.